Teerth Sharma

Performance and Systems Engineer — Compilers, GPU Kernels, ML Infrastructure · Research in Topology

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SUMMARY

Performance and infrastructure engineer working across compiler passes, GPU kernels, inference runtimes and distributed training in **C++**, **Rust** and **CUDA/Triton**. Eleven contributions landed during 2026 in production machine-learning infrastructure at **Google DeepMind**, **Google**, **OpenAI**, **NVIDIA** and **Meta** — each reviewed and accepted by the team that owns the code, one credited by name in the MuJoCo release notes. The through-line is making systems that already run at scale use less memory, run faster, and give the same answer twice. Research alongside: persistent homology applied to attention scheduling and hallucination detection, with one co-authored arXiv preprint.

EXPERIENCE

KarmicSphere Media — NovelPedia (now NovelCult) — Jaipur, India — January 2025 to June 2026

- **Founding Engineer** · January 2025 – December 2025 · **TypeScript, Next.js, Linux, Azure.** Helped build the platform from scratch as part of the founding team — the services behind reading, discovery and the author portal — now carrying **15,000 accounts**, **100,000 viewers** and **200,000 chapters**.
- **TTS Division Lead** · December 2025 – June 2026 · **Python, Django, Azure Speech.** Stood up and led the text-to-speech division, narrating **9,000+ chapters** end to end while holding synthesis to **under $2 of Azure credit per chapter**.

Google DeepMind — Open Source Contributor — 3 merged, July–August 2026

- **MuJoCo** · **C/C++** · **1,281× less solver memory.** Large robot scenes ran out of memory long before they ran out of compute: the constraint solver sized its scratch buffer on the square of the body count. Replacing the dense adjacency matrix and flood fill with a disjoint-set forest made that growth linear, and the union-find primitives are now public API. **Credited by name in the MuJoCo 3.11.0 release notes.** (PR #3396)
- **MuJoCo Warp** · **Python, NVIDIA Warp** · **1.51× faster GPU island discovery in linear memory.** The GPU port carried the same quadratic per-world adjacency plus a DFS stack. Replaced both with a persistent parent array and three allocation-free, CUDA-graph-capturable phases, one 32-lane block per world; island labels stay bitwise-identical under row and endpoint reordering. On a 2,048-world checkpoint with 247,308 active constraint rows: matched the dense kernels exactly in every A-B-B-A block, **1.513× faster by geometric mean, 99% CI [1.406×, 1.584×]**. (PR #1541)
- **MuJoCo** · **C++** · **faster model loading.** Loading a detailed mesh got slower the more detailed it was. Removed a second quadratic scan in convex-hull construction, cutting probes by a factor of 3V/8, with every mesh checksum coming back byte-identical. (PR #3450)

Google — Open Source Contributor — 4 landed, July–August 2026

- **XLA** · **C++** · **deterministic GPU reduction grouping.** Compiling the same module twice could produce different reduction groups: union-find merge order in GroupDisjointReductions followed hash order, so the blockIdx.z constraint in the resulting indexing map varied between builds. Switched to an insertion-ordered set, making group order a function of the graph alone. Landed on main through Google's internal import as commit 3d5df1d. (PR #46539)
- **TensorFlow** · **C++** · **reproducible distributed training.** The same model could schedule differently on two identical runs, which makes a failure impossible to reproduce. The pass that orders collective operations relied on a reachability map that was never transitively closed, so redundant edges survived and hash ordering picked the winners. Rebuilt it as an exact bit-matrix closure, making the emitted graph a function of the model alone. (PR #124410)
- **XNNPACK** · **C** · **32 MiB lower inference workspace.** XNNPACK is the inference backend under TensorFlow Lite, so its memory ceiling decides whether a model fits on the device at all. Taught the planner to reuse the arena gap ahead of the first live tensor: 6.4% off MobileNet V1 peak allocation, outputs byte-identical. (PR #10801)
- **Highway** · **C++ SIMD** · **65× cheaper perfect-hash builds.** Building a perfect hash over a million keys did 894 million pairwise comparisons. Pruning candidates by slice structure took that to 13.6 million and found exactly the same duplicates. (PR #3244)

OpenAI — Open Source Contributor — merged July 2026

- **Triton** · **Triton, Python** · **3.48× faster long-context attention.** Attention over long sequences spends most of its time on blocks that contribute almost nothing. Contributed a sparse attention kernel that skips 81% of blocks at 4K context and still lands within 1e-3 of the dense result, shipped with a benchmark runner so the number survives inspection. Relocated from Triton core on maintainer advice, then merged. (PR #22)

NVIDIA — Open Source Contributor — 2 merged, August 2026

- **NeMo-Relay** · **Rust** · **prompt-cache reuse unblocked.** The cache that should make repeated LLM calls cheap almost never fired. Its governor keyed learning on the first user message, so workflows sharing a system prompt fragmented into one profile per task and never gathered enough evidence to reuse anything. Re-keyed on the stable scaffold and took cross-process agreement from 0.48 to 1.00 — scoped down to a single crate after maintainer review. (PR #481)
- **Topograph** · **Helm, Kubernetes** · **least-privilege RBAC.** The chart granted cluster-wide Kubernetes permissions regardless of configuration — an install that never touches the Kubernetes API still received cluster-wide pods, nodes and daemonsets access. Gated every RBAC rule on the selected engine and provider, dropping the ClusterRole entirely when none applies; 141/141 chart unit tests. (PR #432)

Meta — Open Source Contributor — landed July 2026

- **Pyrefly** · **Rust** · **type-checker crash pinned down.** Meta's Python type checker panicked mid-commit on large codebases when dependency chains ran deeper than the incremental recheck budget. Landed the reproducer that isolates it — 208 chained components — with the maintainer confirming the mechanism on the thread. Shipped in release 1.2.0 via commit 3e90baa. (PR #4180)

SELECTED SYSTEMS

- **Topological ML Toolkit** · **Rust, Python, C++/AVX-512, CUDA.** Turns point clouds and time series into persistence diagrams and Betti features through scikit-learn-style transformers. Four backends, one contract, checked against ripser and GUDHI.
- **Epsilon-Hollow** · **Rust, x86-64 assembly, Linux, QEMU/UEFI, Lean 4.** Research OS testing how much of a kernel can be written in safe Rust: UEFI boot, SMP bring-up, four-level paging, demand paging, scheduling and a VFS — verified under Miri, booted in QEMU on every commit.
- **Caustic** · **Python, PyTorch, HuggingFace.** Detects LLM hallucinations without ground-truth labels by measuring orbit collapse — when a model's internal geometry maps distinct entities onto one answer — and turns that partition into a certified lower bound on error rate, with an inference-time governor that picks its own intervention strength blind. 0.995 AUROC on collapse-type failures at sub-0.5B scale; 163 tests, CI across Python 3.10–3.12, and a published results page documenting where the method does not hold.

RESEARCH

- **OpenCLAW-P2P v7.0** — co-author, arXiv:2604.19792, 2026. Decentralized peer-review protocol for AI research, with resilient persistence and live verification of cited references.

TECHNICAL SKILLS

- **Languages:** Rust, C++, Python, CUDA, Triton, x86-64 assembly (AVX-512), Lean 4, TypeScript
- **Systems:** GPU kernels, compilers, runtimes, operating systems, Linux, QEMU/UEFI, performance profiling, Docker, GitHub Actions, CI/CD
- **Machine learning:** PyTorch, JAX, TensorFlow, scikit-learn, sparse attention, inference optimisation, persistent homology, reproducible benchmarking

EDUCATION

- **Manipal University Jaipur** — 2024 to 2026. Left in 2026 without completing the degree to work full-time. **Open to research apprenticeships, residencies, internships and mentored research placements.**